

ABESEC Ghaziabad

Department of Computer Science & Engineering

SYNOPSIS REPORT

Session 2026–27

Project Title:

[Use the project title already registered with the department]

Project Type:

Application / Research-Oriented Software Project

Name:

[Student Name]

Roll Number:

[Roll Number]

Section:

[Section]

Signature:

Group Member (1):

[Name]

Roll Number:

[Roll Number]

Section:

[Section]

Signature:

Group Member (2):

[Name]

Roll Number:

[Roll Number]

Section:

[Section]

Signature:

Project Guide:

[Guide Name]

Signature:

Remarks:

Date of Submission:

18 August 2026

1.1 Problem Introduction

Recent advances in Large Language Models (LLMs) and generative AI have enabled the development of interactive non-player characters (NPCs) capable of natural-language conversation, role-playing, and context-aware responses. Research such as *Generative Agents* demonstrated that language-model-based agents can maintain memories, reflect on previous experiences, and use those memories to guide future behavior. Subsequent work has explored character-specific role-playing, LLM-driven NPC collaboration, contextualized NPC dialogue, and persistent memory for game characters.

However, recent literature shows that important challenges remain in long-term consistency, memory management, narrative coherence, information control, and evaluation of LLM-driven game characters. A 2026 systematic mapping of 55 studies on AI-generated game dialogue identified memory limitations, repetitiveness, narrative incoherence, and latency as recurring technical problems, while also highlighting the need for stronger and more standardized evaluation methods.

The proposed project addresses these challenges through a small interactive puzzle game consisting of multiple rooms, where each room contains an AI-driven NPC with a distinct personality, relationship state, knowledge set, and interaction strategy. The player must interact with these NPCs, establish sufficient trust or affinity, and obtain appropriate clues to progress through the game.

A central challenge of the project is that NPCs should not possess identical knowledge. Information about previous rooms, player actions, character secrets, and puzzle clues must be selectively available according to its source, scope, relationship, and relevance. Therefore, the proposed system will maintain a centralized game-memory and knowledge repository while generating controlled, NPC-specific knowledge views.

1.1.1 Motivation

Traditional game NPCs commonly rely on fixed dialogue trees and predefined state transitions. Such systems provide strong narrative control but offer limited flexibility when players interact in unexpected ways.

LLM-based NPCs provide more natural and open-ended interaction, but completely unrestricted generation can introduce new problems. NPCs may forget important events, reveal information they should not know, contradict earlier statements, or provide clues inconsistent with the actual game state. Recent user studies also show that greater openness does not automatically improve player experience; unconstrained LLM-NPC interaction can increase perceived autonomy while also increasing cognitive load and reducing usability or trust in some scenarios.

The motivation of this project is therefore to explore a middle ground between:

fixed scripted NPCs
and
unrestricted generative NPCs.

The proposed game will provide NPCs with controlled autonomy, persistent memory, personality, and relationship-aware interaction while retaining deterministic control over important game facts and puzzle states.

1.1.2 Project Objective

The major objectives of the project are:

1. To develop a multi-room puzzle game containing AI-driven NPCs with distinct personalities, goals, and interaction styles.
 2. To design a persistent memory system through which relevant events from earlier parts of the game can influence future NPC behavior.
 3. To develop a centralized knowledge repository containing game events, clues, relationships, NPC information, and world-state information.
 4. To implement hierarchical and access-controlled knowledge views so that each NPC receives only the information relevant to its role, knowledge scope, and relationship with the player.
 5. To develop an affinity/relationship mechanism through which NPC behavior and willingness to provide clues change according to player interactions.
 6. To ensure that AI-generated responses remain grounded in the canonical game state and do not arbitrarily modify puzzle state or reveal unauthorized information.
 7. To evaluate the proposed architecture against simpler NPC architectures using measures such as memory accuracy, information leakage, puzzle completion, consistency, player experience, and system efficiency.
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1.1.3 Scope of the Project

The initial implementation will consist of approximately four to five interconnected rooms. Each room will contain one primary AI-driven NPC with a distinct personality and a different preferred method of establishing rapport with the player.

The player will need to interact with the NPC to obtain information required for solving the room's puzzle and progressing to subsequent rooms. The amount and specificity of assistance provided by an NPC will depend on its relationship state with the player.

The project will maintain a centralized canonical memory/knowledge store. Information stored in this repository will be associated with metadata such as source, timestamp, importance, visibility scope, relationship requirements, and game-state validity. NPCs will not receive the complete database directly; instead, the system will construct filtered knowledge views according to the NPC's authorized information.

The scope of the initial implementation includes:

- 4–5 game rooms
- 4–5 primary NPCs
- distinct NPC personalities
- relationship/affinity management
- event-based memory
- hierarchical knowledge access
- clue generation and controlled disclosure
- persistent cross-room memory
- canonical puzzle/game-state management
- LLM-based NPC interaction
- experimental evaluation

Voice interaction, facial animation, large-scale NPC societies, real-time reinforcement learning, and large open-world environments are outside the scope of the initial prototype but may be explored as future extensions.

1.2 Related Previous Work

Research in AI-driven game characters has progressed from scripted dialogue systems to LLM-based agents capable of memory, planning, social interaction, and contextual dialogue.

Generative Agents introduced an architecture in which agents maintain records of experiences, retrieve relevant memories, form higher-level reflections, and use those memories for planning and behavior. This work provides a major conceptual foundation for persistent interactive characters.

Character-LLM explored character simulation using structured profiles, experiences, and emotional states instead of relying only on simple persona prompts. This demonstrates the importance of explicit character information for consistent role-playing.

Research on **LLM-driven NPC collaboration in Minecraft** investigated how humans interact with GPT-4-driven NPCs during collaborative quests and identified limitations in language-only systems, particularly their lack of rich game-state and visual understanding.

Recent work has also introduced **contextualized generative AI for games**, showing that grounding generated NPC dialogue within gameplay context can improve player presence, autonomy, and enjoyment.

A 2026 systematic mapping of 55 studies on AI-generated game narrative and dialogue identified memory limitations, repetitiveness, narrative incoherence, latency, and weak standardization of evaluation as recurring issues.

Research on **information asymmetry in LLM-agent collaboration** has examined situations in which agents possess different information and must communicate and verify information to solve tasks. These studies are particularly relevant to the proposed game's design, where different NPCs intentionally possess different pieces of information.

Recent work on relationship-aware LLM architectures has also explored representing long-term relational states separately from immediate emotional responses, providing motivation for maintaining explicit relationship information rather than relying only on recent dialogue.

The proposed project builds upon these research directions by combining them within a controlled puzzle-game environment and evaluating how memory organization and knowledge access affect NPC behavior, information disclosure, and puzzle-solving performance.

1.3 Software and Hardware Requirements

Software Requirements

- Operating System: Windows / macOS / Linux
- Game Engine: Unity
- Programming Language: C#
- Backend: Python
- API Framework: FastAPI
- Database: PostgreSQL
- Vector/semantic retrieval: pgvector or equivalent retrieval layer
- LLM API/model: suitable instruction-following language model with structured output/function-calling capability
- Development Environment: Visual Studio Code / Visual Studio
- Version Control: Git and GitHub
- Optional tools: Docker, Postman

Hardware Requirements

- Processor: modern multi-core CPU
- RAM: minimum 8 GB, preferably 16 GB
- Storage: minimum 10 GB free space for development tools and project assets
- GPU: optional for cloud/API-based LLM inference

- Stable Internet connection for external LLM API usage
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1.4 Proposed Method

The proposed development methodology will consist of the following stages.

Step 1 — Game Environment Design

A small puzzle-based game environment containing four to five interconnected rooms will be designed. Each room will contain one principal NPC, a puzzle, relevant objects, and progression clues.

Step 2 — NPC Character Design

Each NPC will be assigned:

- personality traits
- goals
- background
- preferred interaction style
- relationship parameters
- private information
- room-specific knowledge
- clue information

The NPCs will intentionally differ in how they respond to player behavior.

Step 3 — Centralized Knowledge and Memory Store

A central repository will maintain canonical information about:

- game state
- player actions
- NPC actions
- important events
- clues
- relationships
- discovered objects
- room progress
- NPC knowledge

Each information item will include metadata such as scope, source, timestamp, importance, confidence, and access conditions.

Step 4 — NPC Knowledge Views

Rather than giving every NPC access to the full repository, the system will dynamically construct an NPC-specific context based on its allowed knowledge.

For example:

- public information may be available to all relevant NPCs;
- personal information may remain private;
- relationship-specific information may depend on trust or affiliation;
- room-specific information may only be available to characters associated with that room;
- selected high-level events may be propagated to future-room NPCs.

Step 5 — Relationship and Affinity Management

Player interactions will update an explicit relationship state for the NPC. Different NPCs will respond differently to actions such as helping, lying, solving puzzles, showing empathy, following rules, or attempting deception.

Relationship state will influence the NPC's willingness to assist and the specificity of information it provides.

Step 6 — LLM-Based Reasoning

The LLM will receive:

- current game state
- relevant memories
- permitted knowledge
- NPC personality
- relationship state
- current player request

The LLM will generate a structured response or action proposal.

Step 7 — Validation Layer

Generated actions will be checked against the canonical game state before execution. The LLM will not be allowed to directly modify important puzzle state or invent canonical game facts.

Step 8 — Event Logging and Memory Update

Important interactions will be converted into structured events and added to the central repository. Relevant events will update NPC memories, relationships, player-related beliefs, and future knowledge views.

Step 9 — Experimental Evaluation

The final system will be compared with simpler NPC configurations, such as:

- scripted NPCs
- LLM NPCs with persona only
- LLM NPCs with flat conversational memory
- the proposed hierarchical memory and knowledge-access architecture

Performance will be measured using memory accuracy, clue leakage, personality consistency, relationship consistency, puzzle completion, interaction efficiency, player experience, latency, and computational cost.

1.5 Deliverables

The major deliverables of the project will include:

1. A playable four-to-five-room puzzle game.
2. Multiple AI-driven NPCs with distinct personalities and interaction styles.
3. A persistent centralized game-memory and knowledge system.
4. NPC-specific knowledge/access views.
5. Relationship and affinity management.
6. Event-based cross-room memory.
7. LLM-based NPC dialogue and reasoning.
8. Game-state and response validation mechanisms.
9. Experimental benchmark scenarios and evaluation results.
10. Project documentation and source code.
11. Final project report and research-oriented analysis.

Measures of Project Success

The project will be considered successful if:

- NPCs provide contextually appropriate responses;
 - important events remain available across rooms;
 - NPCs do not access unauthorized information;
 - NPC behavior changes appropriately with relationship state;
 - NPCs remain consistent with their assigned personalities;
 - clues remain consistent with canonical game state;
 - players can complete the game through interaction with NPCs;
 - the proposed architecture demonstrates measurable improvements over at least one simpler baseline.
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1.6 Stakeholders

Primary Stakeholders

- Project students and development team
- Project guide
- Department of Computer Science and Engineering
- Faculty evaluators

Secondary Stakeholders

- Game developers
- Researchers working on generative agents and game AI

- Developers of AI-driven NPC systems
- Players interested in interactive narrative games
- Researchers in human-AI interaction and adaptive game systems

The project may serve as a prototype for future systems involving interactive storytelling, educational simulations, serious games, virtual assistants, and persistent social agents.

1.7 Gantt Chart

Activity	Aug-Sep	Oct	Nov	Dec	Jan	Feb	Mar
Literature Review	✓	✓					
Problem Definition & System Design	✓	✓					
Game Environment Development		✓	✓				
NPC Personality & Interaction Design		✓	✓				
Central Memory/Knowledge System			✓	✓			
Relationship & Affinity System			✓	✓			
LLM Integration				✓	✓		
Knowledge Access / Validation Layer				✓	✓		
Game Integration & Testing					✓	✓	
Baseline Implementations					✓	✓	
Experimental Evaluation						✓	✓
Analysis & Documentation						✓	✓
Final Report / Presentation							✓

1.8 References

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